

Section II — Case III:
Defined Contribution Plan and MPF Retirement
Planning

AMA1D03 Introduction to Pension Mathematics

June 15, 2026

(a) Descriptions of the Defined Contribution Plan and Its Application in the Pension System

1. Defined Contribution (DC) versus Defined Benefit (DB)

Under a **defined contribution (DC)** plan, the retirement benefit is not predetermined. The employer and/or employee contribute a specified amount into an individual account, and the final benefit depends on total contributions, investment returns, and fees. Investment and longevity risks are borne primarily by the member.

By contrast, a **defined benefit (DB)** plan promises a benefit formula—for example, a percentage of final salary multiplied by years of service—with the sponsor bearing most investment and longevity risk.

Hong Kong's Mandatory Provident Fund (MPF) System, launched on **1 December 2000**, is a compulsory DC scheme. It forms the **second pillar** of Hong Kong's retirement protection framework:

- **First pillar:** Non-contributory social assistance (e.g., Old Age Living Allowance, Old Age Allowance)
- **Second pillar:** MPF mandatory and voluntary contributions
- **Third pillar:** Personal savings, insurance, and family support

2. Types of MPF Schemes

According to the Mandatory Provident Fund Schemes Authority (MPFA), MPF schemes are classified into three main types:

Table 1: Types of MPF schemes

Scheme Type		Description
Master Schemes	Trust	Open to employees of participating employers and self-employed persons
Employer-sponsored Schemes		Only for employees of a single employer
Industry Schemes		For employees in specific industries (e.g., catering, construction)

Each employee has a personal account within a scheme. When changing jobs, accrued benefits are portable and can be transferred to the new employer's scheme or retained in a personal account.

3. Mandatory and Voluntary Contributions

Mandatory Contributions (MC):

- Employee and employer each contribute **5%** of relevant income
- Relevant income is subject to minimum (HK\$7,100) and maximum (HK\$30,000) monthly thresholds
- Maximum mandatory contribution per party: HK\$1,500 per month

Voluntary Contributions:

- Voluntary contributions (VC) within an MPF scheme
- Tax Deductible Voluntary Contributions (TVC): up to **HK\$60,000 per year** deductible for salaries tax

In Case III, Peter and Mary each contribute HK\$1,000 (5% of HK\$20,000 salary), matched equally by their employers, giving a total monthly contribution of **HK\$2,000**.

4. Benefits, Withdrawal, and Retirement Age

Members can generally withdraw accrued benefits when reaching age **65**. Other early withdrawal conditions include permanent departure from Hong Kong, total incapacity, terminal illness, and early retirement at age 60 with cessation of employment.

Upon withdrawal, members may take a lump sum or convert part of the balance into an annuity through the MPF platform, providing a stream of retirement income.

5. Investment Options, Returns, and Fees

MPF offers multiple fund types: Hong Kong, China, Asia, and global equity funds; mixed assets; bond and guaranteed funds; and index-tracking funds such as the Tracker Fund of Hong Kong.

Members bear **investment risk**. Fund managers charge management fees, typically 0.5%–2% per annum, which reduce net returns over long horizons. In Case III, Peter invests in a Global Fund while Mary invests in a Hong Kong Fund, illustrating how asset allocation affects long-term outcomes under the same contribution structure.

6. Tax Treatment and System Role

Mandatory MPF contributions are tax-exempt up to statutory limits. TVC provides an additional tax incentive for members seeking to close retirement income gaps. Academic and industry studies consistently show that mandatory contributions alone are **insufficient** to maintain pre-retirement living standards, so members are expected to supplement MPF with voluntary savings and, where eligible, government cash benefits.

(b) Compare and Analyze the Defined Contribution Plan — Peter and Mary Examples

1. Modelling Setup

We simulate monthly MPF accumulation from December 2000 until each member reaches age 65 using a Python model (`case3/main.py`). Key assumptions are summarised in Table 2.

Table 2: Key modelling assumptions

Parameter	Value	Note
Start date	1 Dec 2000	MPF launch
Retirement age	65	Normal retirement
Initial monthly salary	HK\$20,000	Case III
Monthly MC (employee + employer)	HK\$2,000	5% + 5%
Salary growth	HK Composite CPI	C&SD annual YoY
Hong Kong Fund proxy	Hang Seng Index (^HSI)	Tracker Fund replicates HSI
Global Fund proxy	Spliced ^GSPC/EFA/ACWI	Full history from Dec 2000
Forward return	4% p.a.	After latest available data
Fund expense ratio	1% p.a.	Typical MPF range
Retirement income	4% withdrawal rule	Annual income from balance

Table 3: Case III participant settings

	Peter	Mary
Start age (Dec 2000)	20	40
Retirement age	65	65
Contribution period	45 years	25 years
Retirement year	2045	2025
Assigned fund	Global (spliced)	Hong Kong (^HSI)

2. Baseline Results

Table 4: Baseline simulation results

Indicator	Peter (Global)	Mary (Hong Kong)
Contribution years	45.1	25.1
Total contributions (HK\$)	1,443,931	723,931
Investment gains (HK\$)	3,552,431	126,930
Final MPF balance (HK\$)	4,996,362	850,862
Final annual salary (HK\$)	555,156	374,107
Annual retirement income (HK\$)	199,854	34,034
Monthly retirement income (HK\$)	16,655	2,836
Replacement ratio	36.0 %	9.1 %

Figure 1 shows Peter’s balance rising steeply over 45 years, while Mary’s balance grows more slowly over 25 years, ending at roughly one-sixth of Peter’s balance. For Peter, investment gains (HK\$3.55 million) far exceed total contributions (HK\$1.44 million), demonstrating the power of long-horizon compounding in a DC system.

Historical Hong Kong equity returns (via the Hang Seng Index) were volatile over Mary’s 2000–2025 contribution period, including the post-2000 downturn and 2008 financial crisis, which helps explain her lower final balance relative to earlier proxy estimates.

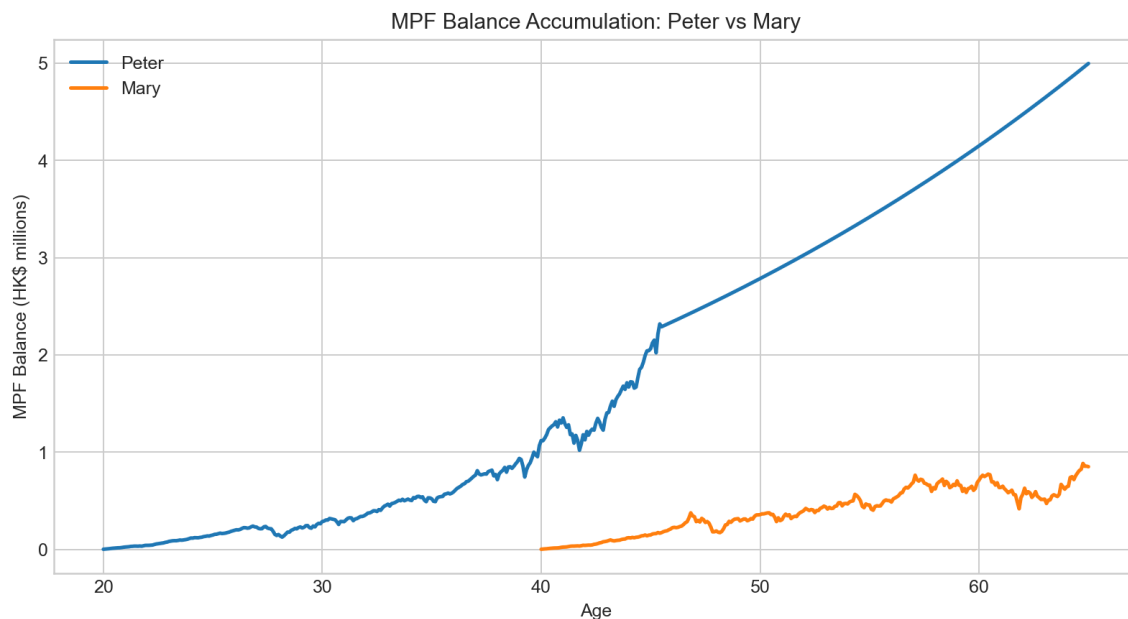


Figure 1: MPF balance accumulation: Peter versus Mary

3. Advantages and Disadvantages

Advantages:

1. **Employer matching:** The employer's 5% contribution doubles savings without additional employee cost.
2. **Portability:** Accrued benefits follow the employee across jobs.
3. **Investment choice:** Members can select funds aligned with risk tolerance.
4. **Tax incentives:** TVC of up to HK\$60,000 per year is tax-deductible.
5. **Compounding:** Early starters benefit enormously from decades of compound growth.

Disadvantages:

1. **Investment risk:** Members bear full downside risk; Mary's Hong Kong fund underperformed the global fund over the sample period.
2. **Low replacement ratio:** Even Peter achieves only 36% replacement from mandatory contributions, below the 60–70% commonly recommended.
3. **Fee drag:** Management fees compound negatively over 25–45 years.
4. **No inflation guarantee:** Retirement income depends on uncertain market returns.
5. **Late starter penalty:** Mary's 9.1% replacement ratio shows that starting at 40 cannot be fully offset by fund selection.

4. Start Age versus Fund Choice

Table 5 reports replacement ratios by start age. Table 6 reports fund-swap sensitivity at fixed start ages.

Table 5: Replacement ratio by start age and fund

Start Age	Fund	Final Balance (HK\$)	Replacement Ratio
20	Global	4,996,362	36.0%
20	Hong Kong	2,427,886	17.5%
30	Global	3,417,078	30.0%
30	Hong Kong	1,500,213	13.2%
40	Global	2,063,587	22.1%
40	Hong Kong	850,862	9.1%
50	Global	520,922	6.9%
50	Hong Kong	457,617	6.0%

Table 6: Fund-swap sensitivity

Name	Assigned RR	Swapped RR	Change (pp)
Peter (age 20)	36.0%	17.5%	−18.5
Mary (age 40)	9.1%	22.1%	+13.0

Key finding: Comparing start ages at the same fund shows a **27 percentage-point gap** (36% versus 9.1% for global and Hong Kong funds at ages 20 and 40). Peter at age 20 with the under-performing Hong Kong fund (17.5%) still ends with a far higher balance than Mary. **Contribution horizon dominates fund selection.**

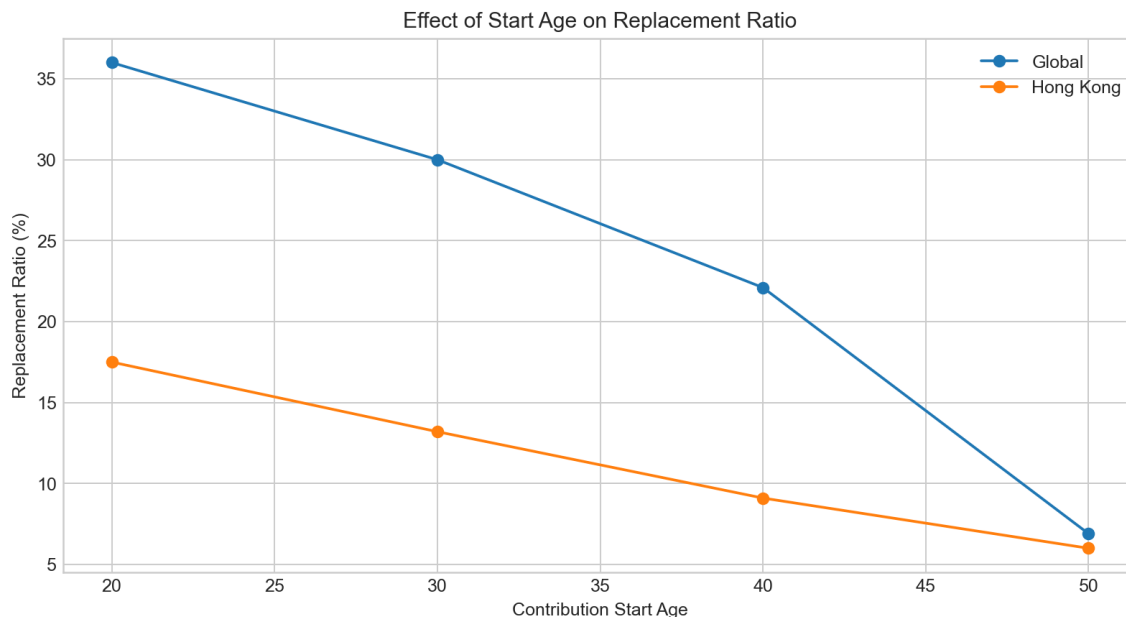


Figure 2: Effect of start age on replacement ratio

5. Living Expenses, Inflation, and Extra Contributions

Financial planners often target a **60–70% replacement ratio**. Peter’s MPF income leaves a monthly gap of approximately HK\$11,360 to reach 60%; Mary’s gap is approximately HK\$15,269 despite a lower final salary.

Salaries and mandatory contributions are indexed to CPI, but real purchasing power of retirement income depends on whether investment returns exceed inflation. A 1% fee differential over 45 years can materially reduce the final balance.

Table 7: TVC required to reach target replacement ratio

Member	Current RR	Target RR	Extra Monthly TVC (HK\$)
Peter	36.0%	60%	1,582
Peter	36.0%	70%	2,242
Mary	9.1%	60%	13,205
Mary	9.1%	70%	15,799

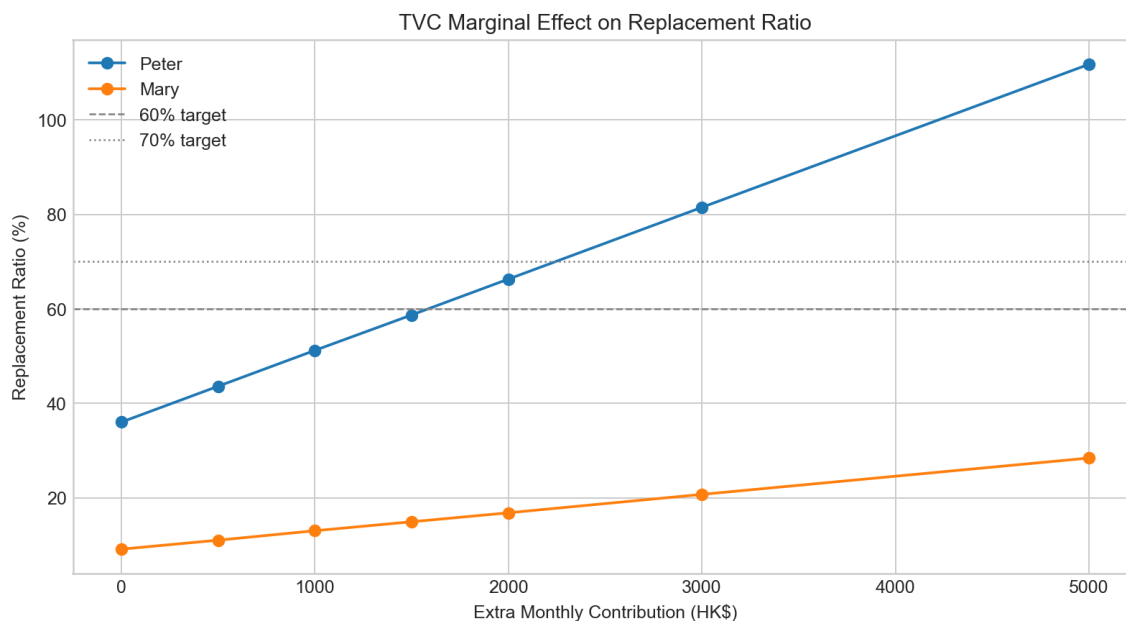


Figure 3: Marginal effect of TVC on replacement ratio

Mary urgently needs substantial TVC or other income sources. Peter still requires about HK\$1,600 per month extra to reach 60% replacement, achievable within the TVC tax-deductible limit if planned annually.

(c) Supporting Retirement Under DC with the Social Security System

1. Hong Kong Three-Pillar Framework

Unlike Canada's CPP or US Social Security, Hong Kong has **no universal earnings-related public pension**. Government support is means-tested:

- **First pillar:** OALA, Old Age Allowance, CSSA
- **Second pillar:** MPF mandatory and voluntary (TVC)
- **Third pillar:** Private savings, insurance, family support

2. Old Age Living Allowance (OALA)

Eligible Hong Kong residents aged 65+ may receive Normal OALA (HK\$2,065/month) or Higher OALA (HK\$4,185/month) in 2025/26, subject to income and asset tests. For a single person, the asset limit for Higher OALA is approximately HK\$401,000. **MPF accrued benefits not yet withdrawn may be counted as assets**, potentially disqualifying members with substantial balances.

3. Combined Retirement Income

Table 8: Retirement income breakdown (selected scenarios)

Member	Scenario	MPF	OALA	Total	Combined RR
Peter	No OALA	16,655	0	16,655	36.0%
Peter	Normal OALA	16,655	2,065	18,720	40.5%
Peter	Higher OALA	16,655	4,185	20,840	45.0%
Mary	No OALA	2,836	0	2,836	9.1%
Mary	Normal OALA	2,836	2,065	4,901	15.7%
Mary	Higher OALA	2,836	4,185	7,021	22.5%

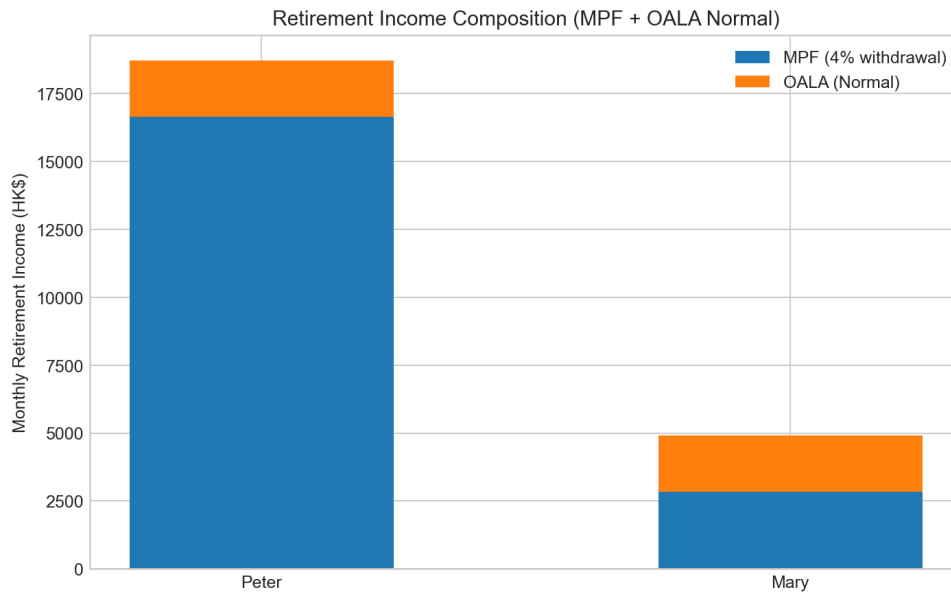


Figure 4: Retirement income composition: MPF plus Normal OALA

4. Illustrative Strategies

Peter (retirement 2045):

- MPF provides a solid base (36% replacement ratio).
- Add TVC of about HK\$1,600 per month to reach 60% replacement from MPF alone.
- OALA is unlikely given expected asset levels; the plan should not rely on it.
- Consider MPF annuitisation for longevity protection.

Mary (retirement 2025):

- Mandatory MPF alone is severely inadequate (9.1% replacement ratio).

- Higher OALA (combined RR 22.5%) still leaves a large gap to 60%.
- Begin TVC as early as possible; consider working beyond 65.
- Build third-pillar savings given limited compounding time.

A realistic mixed plan for Mary—MPF (HK\$2,836) plus Higher OALA (HK\$4,185) plus part-time work or savings (HK\$5,000)—yields about HK\$11,021 per month, still below the 60% target of HK\$18,706.

5. Conclusions on Extra Contributions

Table 9: Summary recommendations

Question	Peter	Mary
Is mandatory MPF sufficient?	No (36%)	No (9.1%)
Should they add TVC?	Yes, moderate	Yes, urgent
Can social security fill the gap?	Unlikely	Partially only

Hong Kong’s DC system shifts retirement responsibility to the individual. Early, consistent contributions—as in Peter’s case—are the most powerful lever. Social security (OALA) provides a safety net for lower-asset retirees but cannot substitute for inadequate lifetime savings. **Both Peter and Mary should make extra contributions; the urgency and required amount are far greater for Mary.**

Key Formulae

Future value with CPI-linked contributions:

$$FV = \sum_{t=1}^n PMT_0(1+g)^{t-1} \prod_{k=t+1}^n (1+r_k) \quad (1)$$

Replacement ratio with a 4% withdrawal rate:

$$RR = \frac{FV \times 0.04}{Salary_{final}} \quad (2)$$

Monthly TVC gap:

$$Gap_{monthly} = Target_{monthly} - \frac{FV_{MPF} \times 0.04}{12} - OALA_{monthly} \quad (3)$$

References

- [1] Mandatory Provident Fund Schemes Authority, *MPF Fund Performance Platform*, <https://mfp.mpfa.org.hk/>.

- [2] Social Welfare Department, *Old Age Living Allowance*, <https://www.swd.gov.hk/oala/>.
- [3] Census and Statistics Department, *Composite Consumer Price Index*.
- [4] Investopedia, *Individual Retirement Account (IRA)*, <https://www.investopedia.com/terms/i/ira.asp>.

Appendix: Simulation Reproducibility

All numerical results are generated by the Python pipeline in `case3/`. To reproduce tables and figures:

```
cd case3
python3 main.py
python3 src/validate.py
```

Output files are stored in `case3/output/`. Assumptions are documented in `case3/data/assumptions.md`.